MIL-P-12001D
29 March 1977
SUPERSEDING
MIL-P-12001C
21 August 1967

MILITARY SPECIFICATION

PROBE LIGHT KIT, INSPECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a probe light kit with a battery-powered illumination device, a viewing mirror, a magnetic retriever for ferrous objects, and a carrying case.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

J-C-580	- Cord, Flexible, and Wire, Fixture, (Electrical, 0- to 600-Volt Service).
QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6230

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-B-18/9	- Battery, Dry, BA-30.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-2312	- Felt, Hair or Wool: Mildew Resistant, and Moisture Resistant Treatment for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The probe light kit shall consist of a light, one master extension arm, three regular extension arms, mirrors as specified herein, a magnet, and a carrying case.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Plating. The following components and parts shall be chromium plated in accordance with QQ-C-320, Type I, Class 1:

(a) Tubes for extension arms.
(b) Forms and protective shields.
(c) Ball-and-socket joint fittings.
(d) Hinges and exposed hardware.

3.3 Weight. The probe light kit, without batteries, shall weigh not more than 2-1/2 pounds.

3.4 Operation. The assembled probe light connected to batteries shall produce a forward-directed beam of illumination.

3.5 Components.

3.5.1 Extension arms. The master and three regular extension arms shall be fabricated from rigid metal tubing having an outside diameter of 0.25 inch and a length of 6 inches, plus or minus 0.5 inch. The arms shall be electrically wired internally and connections shall be provided

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3.6 Description. The light shall be in accordance with W-L-[illegible] and as described herein.

to operate the lamp adapter assembly (see 3.5.3) using either the master extension arm or the master extension arm with any combination of regular extension arms. All electrical connections shall be secured, soldered, and insulated in accordance with current commercial practice. Threads shall be national fine series thread conforming to NBS Handbook H28.

3.5.1.1 Master extension arm. The master extension arm shall be threaded internally at one end to receive the inspection lamp or a regular extension arm. The other end shall be provided with a quick-connecting, positive-locking device for attaching to the socket of the extension cord (see 3.5.2).

3.5.1.2 Regular extension arms. The regular extension arms shall be threaded internally at one end and externally at the other end to mate with the internal threads of the master extension arms or with a regular extension arm. The regular extension arms shall be interchangeable.

3.5.2 Electrical extension cord. One electrical extension cord not less than 4 feet in length, consisting of No. 18 AWG, 2-conductor cable conforming to J-C-580, Type SJO, shall be provided. One end of the cable shall be provided with an insulated tip plug on each conductor to connect to the tip jacks on the carrying case (see 3.6). The other end of the cable shall be provided with a quick-connecting, positive-locking-type socket for attaching to the master extension arm (see 3.5.1.1).

3.5.3 Lamp adapter assembly. One lamp adapter assembly consisting of metal housing threaded at one end with national fine series threads in accordance with NBS Handbook H28 for use with any extension arm and a removable clear glass lens at the other end shall be provided. The adapter shall be threaded on the interior to accommodate a No. 1768 lamp and shall provide for a continuous electrical connection from the lamp to any extension arm. One lamp shall be in the adapter, and one replacement lamp shall be in a fitted compartment in the case.

3.5.4 Mirror assembly. One 1-inch-diameter mirror mounted in a steel frame shall be provided. A double ball-and-socket arrangement shall connect the mirror mounting frame to a 6-inch, plus or minus 0.5 inch, steel rod that is fitted with a clamping device to securely attach the rod and mirror to any of the extension arms. When specified (see 6.2), a magnifying mirror 3.5 inches in diameter with a 5-inch focal point and a magnification of 2-1/2 times shall be furnished in addition to the 1-inch mirror.

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3.5.5 Magnetic retriever. One permanent-type magnetic retriever not more than 0.25 inch in diameter which shall lift and hold a weight of not less than 2 ounces shall be provided. A clip device shall be provided to attach the retriever to an extension arm.

3.6 Carrying case. A wooden carrying case with a hinged lid and a snap fastener shall be provided. The overall dimensions of the case with the lid closed shall not exceed 10 inches in length, 4.5 inches in width, and 3 inches in depth. A leather or nylon carrying strap not less than 3 feet in length shall be provided. The carrying strap shall be attached on both sides of the case near one end. The case shall hold all components specified in 3.5, plus six Size D dry batteries conforming to MIL-B-18/9. Components shall be secured to the case by holddown devices. Components shall be protected from damage due to rough handling by the use of felt conforming to MIL-F-2312, Type III. A battery compartment shall be provided and shall have a complete series electrical connection from the batteries through an "ON-OFF" rheostat to two tip jacks for attachment of the extension cord. The rheostat control knob and tip jacks shall be located on the outside of the carrying case. Batteries shall not be furnished with the kit.

3.7 Identification marking. The kit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the kit on an identification plate conforming to MIL-P-514, Type I, Style 3, Composition C.

3.8 Treatment and painting. All parts of the probe light kit normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A.

3.9 Workmanship. The probe light kit shall be free from defects such as rust, cracks, burrs, roughness, and other defects that could impair its operation or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective. Batteries required for testing shall be furnished by the contractor and shall be at full voltage.

4.4 Inspection procedure.

4.4.1 Examination. The probe lights and cases shall be examined as specified herein for the following defects:

Major

101. Material or plating not as specified.
102. Maximum weight limit not as specified.
103. Components or parts missing or not as specified.
104. Carrying case not as specified.
105. Workmanship not as specified.

Minor

201. Identification marking not as specified.
202. Treatment and painting not as specified.

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4.4.2 Tests.

4.4.2.1 Magnetic retriever. The magnetic retriever shall be assembled on the master extension arm. Failure of the magnetic retriever to lift and hold a 2-ounce weight for at least 5 minutes shall constitute failure of this test.

4.4.2.2 Light. The kit components including all extension arms shall be assembled. The rheostat shall be actuated from the "OFF" position to the "ON" position and then from low intensity to maximum intensity. The actuation of the rheostat shall be such that the graduations in light intensity are clearly distinguishable. This test shall be repeated 10 times. Failure of the light to operate, evidence of faulty electrical connections or contacts, inability of the rheostat to control the intensity of the light throughout the entire light range, or inability of the rheostat to turn off the lamp shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major and minor defects. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

Major

106. Materials or containers not as specified. Each incorrect material or container shall be considered one defect.
107. Box closure not Method V as specified.
108. Strapping not as specified.

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Minor

203. Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.1.1 Level A. Each carrying case, containing the components of the kit, including the technical publications, shall be preserved in a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.1.1 Repair parts and tools. Repair parts and tools shall be preserved in accordance with the criteria and applicable method of preservation of MIL-P-116.

5.1.2 Level B. Probe light kits, repair parts, and tools shall be preserved as specified for Level A, except that boxes shall be Type CF, Class Domestic, Variety SW, Grade 125, and Method V closure shall not be required.

5.1.3 Commercial. Probe light kits, with technical publications, repair parts, and tools shall be preserved to afford protection against deterioration and damage during shipment and handling from the contractor to the initial destination.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Probe light kits, repair parts, and tools preserved as specified in 5.1 shall be packed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Probe light kits, repair parts, and tools shall be packed as specified for Level A, except that boxes shall be Domestic Type and strapping shall be Finish A.

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5.2.3 Commercial. Probe light kits, repair parts, and tools shall be packed to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Commercial packaging marking shall reflect as a minimum, the item name, national stock number, quantity, contract or purchase order number, and appropriate address markings. The marking shall be applied with a waterproof material, shall be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The probe light inspection kit is intended to be used to inspect small bores, recesses, and inaccessible areas in castings, machines, and machine parts and to retrieve small ferrous metal objects.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a magnifying mirror is required (see 3.5.4).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When other than Finish B strapping is required (see 5.2.1).

6.3 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each probe light kit.

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the probe light kits.

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User activities:
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Preparing activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-P-12001D Probe Light Kit, Inspection

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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